

SEAN HERLIHY

Los Angeles, California | 4073739370 | sherlihy@usc.edu | <https://www.linkedin.com/in/sean-herlihy-5318008/>

EDUCATION

University of Southern California

Los Angeles, California

Master of Science, Computer Science

Expected May 2026

- Relevant Coursework: Data Structures & Algorithms, Distributed Systems, Machine Learning, Web Technologies, Information Retrieval

University of Central Florida

Orlando, Florida

Bachelor of Science, Computer Science

August 2017-February 2024

- Graduating GPA: 3.65/4.00
- Focus: Software Engineering, Machine Learning

EXPERIENCE

Cambridge Savings Bank

Boston, Massachusetts

Software Engineering Intern, Residential Lending

May 2025-August 2025

- Designed and implemented data processing and reporting pipelines, improving workflow efficiency and reducing manual effort by 3+ hours/month
- Analyzed thousands of records within large-scale financial datasets using Python and SQL, identifying trends and improving internal reporting accuracy
- Collaborated with cross-functional teams to design and prototype AI-driven software solutions, contributing to internal system improvements
- Developed data visualizations and dashboards to support business decision-making

Royal Custom Builders Inc.

Winter Garden, Florida

Financial Controller

December 2023-May 2024

- Developed data-driven budgeting and reporting tools, improving cost tracking across multiple projects
- Automated financial workflows and improved data consistency across systems
- Integrated historical project data into reusable models for forecasting and planning

Expert Painting & Pressure Washing

Orlando, Florida

Tech Consultant

May 2023-August 2023

- Built and deployed a custom full-stack scheduling and management system, reducing operational workload by 13 hours/week
- Designed backend logic and data models to support scalable scheduling and task management
- Improved system reliability and reduced scheduling errors through automation
- Delivered user training and documentation to ensure successful system adoption

PROJECTS

Epilapse

August 2022-December 2022

- Engineered and deployed a full-stack application using React, Node.js, MongoDB, and REST APIs on AWS
- Implemented real-time communication using WebSockets for live health tracking
- Designed and optimized backend APIs and database interactions to improve performance and support scalable workloads
- Collaborated in Agile sprints with a team of 5, using Git for version control

Lead Infrastructure Engineer, NASA Simulation Exploration Experience

August 2022-May 2023

- Architected and deployed a secure VPN-based distributed system supporting hundreds of concurrent users
- Designed system infrastructure to ensure scalability, fault tolerance, and network reliability
- Developed and deployed interactive simulation components

SKILLS

Languages: Python, Java, C++, C#, JavaScript, TypeScript, SQL

Backend & Systems: Node.js, Express.js, RESTful API Design, WebSockets, Distributed Systems, Microservices, Asynchronous Programming

Frontend: React, Angular, HTML, CSS

Databases: MySQL, MongoDB, Database Design, Query Optimization

Cloud & DevOps: AWS (EC2, S3), Docker, CI/CD

Data & ML: Pandas, NumPy, Scikit-learn, Data Analysis, TensorFlow

Tools: Git, GitHub, Linux, Tableau, Excel

Practices: Data Structures & Algorithms, OOP, System Design, Testing (JUnit), Agile/Scrum